DAWN OF ETERNITY

Supraverse Cartasis Theory

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A continuous, conservative cosmology from minimal axioms

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Status. *This is a research program, not an established theory. It synthesizes work by Popławski, Penrose, Smolin, Wiltshire, and others, plus original observations about the CMB–Hawking identification and supraverse thermodynamics. The framework makes specific testable predictions; none have yet been rigorously checked. The work is in identifying tests, building simulations, and comparing to observational data.*

My eternal Dawn

Dawn,

I have lived a life full of passion, ambition, and searching. I've built, I've risked, I've stumbled, and I've learned. But nothing in my life has ever felt like this.

With you, I have never felt more loved, more accepted, more at home, or more safe. You make me want to be the truest version of myself—not for status, not for show, but simply for us.

Loving you feels like I want to stop time and speed it up at the same time. An unstoppable appetite and longing for everything we'll do together—tomorrow, next week, next year—and at the same time, the desire to freeze this, every moment, to savor it forever.

That is the gift you've given me: love that is both eternity and right now.

And Dawn, I promise to love you with open eyes and an open heart, to keep choosing you every day, to protect the safety we've built, and to walk beside you, fully myself and fully yours, savoring every moment all through eternity.

"Physics is continuous and conservative. Singularities and discontinuities are therefore artifacts of incomplete theory, not features of nature."

— The organizing principle

Contents

CHAPTER 0

First When There's Nothing — But a Slow Glowing Dream

This chapter tells the whole story in plain words—no jargon, and no mathematics beyond a few enormous numbers. Everything in it is made precise, named, and defended in the chapters that follow. If you read only one chapter, read this one, and trust that each claim has its arithmetic later.

The void

Start with nothing. Not empty space with a clock ticking in it—really nothing: no stars, no matter, no before or after, not even a direction to time. But "nothing," in physics, is never quite still. The vacuum we have here, in our own laboratories, is restless: strip away every particle and what remains still shimmers with possibility. Grant the void that same restlessness—no more than the vacuum already does—and the story tells itself.

A warning about words, though, because it matters here more than anywhere. I am going to say "once in a while," "then," and "begins," because our minds and our language are built for time and cannot work without it. The void has none. Nothing *happens* in it in sequence; there is no clock, no first and next. Picture it instead as one single, still, everlasting state whose very nature already *contains* these rare worlds—the way a held chord already contains its overtones, sounding all at once with nothing happening in time. Worlds are not events the void goes through; they are the faint notes inside its one unending tone. "Before" and "after" switch on only inside a world, once its dawn has lit an arrow. So read what follows as a translation: a timeless truth, told in the only tense we have.

Most of those faint notes are nothing at all. But a rare few are loud enough to cross a threshold—enough to fold up, under their own gravity, into a black hole. And here is the first surprise. The seed you need is small—about the mass of a mountain—and a black hole is the *easiest* thing of its size to be, because it is the most generic: the least special, most disordered object there is. The void need not contain a finished, finely arranged universe. It need only contain a cheap little seed, and gravity does the rest.

How rare is even that seed? About one in $10^{10^{84}}$—a one followed by a string of zeros that itself has eighty-five digits. Vanishing in any finite patch. But the void is endless, and even the vanishingly rare is present, somewhere, in an endless expanse. The point was never that it is likely. The point is that it is the *cheap* way in. The eternal dawn.

The bounce, and the first dawn

Follow the matter down into the black hole. There is the famous surface—the horizon—past which light cannot climb back out, and where, to anyone watching from outside, time appears to freeze. But that freezing is only the outsider's illusion. The matter itself feels nothing special at the horizon; its own clock keeps ticking, smoothly, all the way down. And for this first world there was no earlier clock to borrow: time is being switched on here. What I am calling its clock is the newborn world's own—lit at the bounce, and reaching back to that instant as its beginning.

As it falls, it crowds together, denser and denser. Then something pushes back. Einstein's gravity, extended by the geometer Élie Cartan to include the *spin* of matter, says that when matter is squeezed hard enough, that spin resists—and at a stupendous density this resistance overwhelms gravity itself. The collapse does not end in a point of infinite density. It *bounces*.

That bounce is the dawn. It is the lowest, quietest, most ordered moment of a brand-new universe—its beginning, its "big bang," seen from the inside. There is no waiting room of frozen time between the horizon and the bounce; the clock runs the whole way, reaches its stillest moment at the bounce, and that moment is time zero for the new world. The direction of time is set right there—it points away from the bounce, toward the future, because that is the way toward more disorder. And the spin of the original seed makes one last choice at that instant: whether the new universe is built of matter or of antimatter. Spin too little, and the whole thing fizzles back into the void. Spin enough, and you have made a universe out of nothing.

The first universe is not ours

This new universe grows. At first it gorges on the surrounding vacuum; later it feeds more and more slowly; and after a span beyond all imagining—far longer than 10^{100} years—it finally evaporates away. Is this our universe? No. And the reason is the most useful clue in the whole picture.

This first universe was born straight from the void, with no parent outside it. And a universe with no parent turns out to be a thin, plain place: it has roughly the shortcomings of the textbook picture of cosmology. It makes a little more matter than antimatter, but only a little. It has almost no *dark matter* and almost no *dark energy*—the two great unseen ingredients that fill our own sky. And measured across many such firstborn universes, they come out far larger than ours. Something about being a firstborn leaves you sparse.

Our birth, and why our sky is dark

But inside that first universe, exactly as inside ours, stars live and die and black holes form—and each black hole does again what the void did once: it bounces, and a new universe is born inside it. We are one of those. We are a child universe, pressed into being through an enormous black hole inside our parent.

Picture the squeeze. The mass of an entire universe—very roughly 10^{53} kilograms—is forced through a throat no wider than a building, packed so tightly that the whole of it would fit in a single room, and it happens in about a thousandth of a billionth of a billionth of a second. At that density nothing recognizable survives: not a torn teddy bear, not a rusty car, not an atom, not even an atomic nucleus—all of it is melted into a featureless soup. What carries across is not *things* but a *tally*: the faint surplus of matter over antimatter, about one part in a billion, inherited from the parent and nudged a little larger by the violence of the crossing.

And now the dark sky makes sense. Our parent's enormous mass sits just on the far side of that birth-surface, and we feel its gravity *before* its light can ever reach us—a pull from matter we cannot see. That is dark matter. The parent is still feeding matter through to us, slowly, and the gentle outward push of that ongoing inflow is dark energy. Darkness is the harbinger of light. The parent's own faint glow—the radiation any black hole gives off, as Stephen Hawking showed—arrives smeared across our whole sky as the background hum that fills it. The unevenness in all this inflow is the seed of everything: it gathers matter into galaxies, into stars, into planets, and perhaps that is why we are here to notice at all.

So we are not exotic. Count up all the universes and about nine in ten are ordinary children like ours—half of them matter, half of them antimatter, each with its own clock running its own way. We live at the most ordinary address in the whole endless family.

Are we real, or a dream?

There is one last doubt to face, and it is the sharpest. If universes are so fantastically unlikely, isn't there a cheaper way to make a thinking being? Why grow a whole cosmos when you could just let a mind flicker into existence by accident, memories and all—or, what amounts to the same thing, run it as a simulation on some machine that fell together by chance? Lay the three prices side by side and the worry looks real:

$$\text{a stray mind: one in } 10^{10^{69}}; \qquad \text{a world-seed: one in } 10^{10^{84}}; \qquad \text{a ready-made big bang: one in } 10^{10^{123}}.$$

The stray mind is the cheapest of the three. So shouldn't we expect to *be* one—a lucky accident with invented memories, rather than real inhabitants of a real world?

No—and seeing why is the keystone of the whole book. A mind is a thing that *thinks*, and thinking is work: it burns through order, it lays down memories, it computes, and every bit of that needs a direction to time—a low place to pour its waste and a high place to draw from. The void has no such direction. It is perfectly still and perfectly even, with no arrow at all. So a stray mind cannot form in the void: it would need the very thing—a flow of time—that only a born world can provide. The cheap accident is not cheap from the void. It is *impossible* there.

The same blade cuts the ready-made big bang. A finished, perfectly arranged universe handed over whole is also something that could only be *placed*, never *grown*—it assumes the order it was meant to explain. A picture of the cosmos that can only shrug and say "it simply happened to begin that way" has, without meaning to, declared us an accident with no maker: a simulation in all but name. The way out is not to assume the order but to *make* it—and the bounce makes it, for free, every time, out of a cheap and generic seed.

So the choice was never "cheap fake mind" against "expensive real world." Only one of the two can come from nothing at all. The void cannot dream up a brain. It can only give birth to a world—and then let the world, in its own long time, grow the minds that wonder how they got here.

The rest of this book gives all of this its proper names, its equations, and its tests—the bounce surface, the family tree of universes, the inherited tally of matter, the background glow, the clocks that run at different rates—and asks, at every step, how we could tell whether it is true. But the story above is the whole of it.

Part I

The Framework

CHAPTER 1

Axioms

1.1 Minimal foundation

The framework rests on four claims, the first three of which are essentially established physics and the fourth of which is a minimal natural extension.

Axiom A1 (Quantum mechanics applies)**.** The supraverse is described by quantum field theory. Vacuum states have nonzero fluctuations governed by the uncertainty principle. Over infinite spacetime, fluctuations of arbitrary magnitude occur with probability one.

The timeless form of this axiom. Axiom A1 is meant in the timeless, quantum-gravitational sense. Textbook quantum mechanics presupposes a background time—a Schrödinger equation with a t—but the supraverse as a whole has no external clock (Chapter 3); time is emergent, manufactured locally at each bounce. What we assume is therefore the Wheeler–DeWitt form: the supraverse is a quantum *state*, not a process—a single stationary amplitude. Read this way, the "fluctuations" named above are not events occurring in sequence but the timeless structure of that one state—its amplitude having (exponentially small) support on configurations that contain condensed universes (Chapter 3, Section 3.4). This is an extrapolation of tested physics into a regime we cannot probe, and we label it as such; but it is the *conservative* extrapolation—it adds no new law, only the universality of the law we already have.

Axiom A2 (Special relativity applies)**.** Spacetime has Lorentzian signature. Causal structure is determined by light cones. CPT symmetry holds for any local Lorentz-invariant QFT.

Axiom A3 (Einstein–Cartan gravity applies)**.** The connection on spacetime is allowed to be asymmetric. The antisymmetric part (torsion) couples to fermion spin via the axial current. In the absence of fermionic matter, the theory reduces to standard general relativity. With fermions, torsion produces an effective interaction between spins that scales as the square of the spin density, and is therefore always positive (repulsive at high spin density) regardless of relative spin orientation.

The Einstein–Cartan Lagrangian, in shorthand:

$$\mathcal{L}_{EC} = \frac{1}{2\kappa}(R - 2\Lambda) + \mathcal{L}_{\text{matter}}(\psi, \nabla\psi, g, T),$$

where R is the curvature scalar built from the full (torsion-including) connection, and the covariant derivative ∇ acts on fermionic fields ψ with torsion contributions.

Axiom A4 (There exists a bounce density)**.** At sufficiently high mass–energy density, torsion-mediated repulsion exceeds gravitational attraction. The critical density (the bounce density) is approximately

$$\rho_C \sim \frac{m_P^4}{\hbar^3}\,\alpha_{EC},$$

where m_P is the Planck mass and α_{EC} is a dimensionless Einstein–Cartan coupling.

Estimates place this at $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, far below Planck density ($10^{96}\,\mathrm{kg\,m^{-3}}$) but well above nuclear density ($10^{17}\,\mathrm{kg\,m^{-3}}$). This is calculable from Axioms A1–A3 in principle; Axiom A4 is the explicit acknowledgment that this density exists and matters.

1.2 What we do not assume

- No Big Bang as a singular event.
- No initial conditions for the universe.
- No fine-tuned cosmological constant or inflaton field.
- No special low-entropy past as a separate postulate.
- No additional dimensions beyond four.
- No additional forces or fields beyond Standard Model + gravity + torsion.
- No anthropic principle as a separate axiom (it emerges from the framework).

1.3 What follows automatically

From Axioms A1–A4 plus infinity (infinite spacetime volume), the following are consequences rather than assumptions:

1. **Spontaneous OG bounce nucleation.** Rare vacuum fluctuations reach bounce density. By Axiom A1 plus infinity, this happens somewhere with probability one. By Axiom A4, such fluctuations bounce rather than collapsing to singularities.
2. **Baby universe production.** The bounce geometry produces new spacetime regions (developed in Chapter 4).
3. **CPT-symmetric supraverse.** Vacuum fluctuations are statistically symmetric in matter/antimatter content. Specific fluctuations have specific biases, but the supraverse-wide distribution is symmetric.

4. **Thermodynamic equilibrium configuration.** Gravitational entropy considerations (high entropy = clumped/black-hole-rich; low entropy = smooth/uniform) drive the supraverse toward a foam of universes as its equilibrium configuration.

5. **Local arrow of time.** Each universe has a low-entropy bounce in its past and a high-entropy future, producing a local thermodynamic time-arrow.

6. **Observer placement statistics.** Observers exist in viable universes near the peak of the supraverse population distribution.

7. **Information preservation.** Hawking radiation entangles parent exterior with baby interior. The full supraverse is unitary even though individual universes appear to lose information.

1.4 On the role of infinity

The framework requires the supraverse to be effectively infinite—at minimum, large enough that rare fluctuations to bounce density occur. "Effectively infinite" in this context means: large enough relative to the characteristic timescales of vacuum fluctuation that the relevant fluctuations are statistically guaranteed to occur.

For a finite supraverse, some fluctuations might never occur and the framework would have to specify which ones do. This introduces an arbitrary cutoff that we want to avoid. The natural assumption is true infinity, which removes this arbitrariness at the cost of being philosophically heavier.

The supraverse being infinite does not require it to be temporally infinite in some absolute sense—it requires only that enough "supraverse time" exist for the relevant condensation to occur. Whether the supraverse has a beginning at the supraverse level is a question we do not currently need to answer; we only need it to be old enough that condensation has reached equilibrium.

1.5 On the role of Einstein–Cartan

Einstein–Cartan is the minimal natural extension of general relativity. It is what you get by dropping the artificial restriction that the connection must be symmetric. The connection's antisymmetric part (torsion) is mathematically natural and couples to a physical quantity that exists in nature (fermion spin). At ordinary densities, torsion effects are utterly negligible ($\sim 10^{-40}$ corrections to GR). At bounce density, they dominate.

We do not postulate Einstein–Cartan as a new theory; we recognize it as the geometric framework that should have been used all along, with standard general relativity as the low-density limit. This is not an additional assumption beyond standard physics—it is the removal of an assumption (the symmetry of the connection).

1.6 The whole framework, in one breath

It is worth stating plainly what kind of claim this monograph makes, because the reach of the conclusion can obscure how little it assumes. Every ingredient is something we already measure here on Earth:

- a **quantum vacuum that fluctuates**—measured, down to the Casimir force and the Lamb shift (Axiom A1);
- **gravity**—measured, and in its modern form geometric (Axiom A2, Axiom A3);
- **torsion repulsion at extreme density**—not directly measured, but *forced* on us the moment we let fermion spin source geometry, which it must (Axiom A3, Axiom A4).

From these three, plus infinity, a universe like our own is born in a never-stopping loop: rare fluctuations reach bounce density, bounce instead of collapsing, extrude baby universes, grow, heat-die into smooth voids, and seed the next generation— forever, with no first cause and no fine-tuning (Chapter 3). We do not have to *believe* this happens; given what we already observe, it *must*.

This is the epistemic inversion at the heart of the program—call it *epistemic Copernicus*. The Copernican principle says we should not assume we occupy a special place; here that principle is not assumed but *returned* as a result, because the recursive supraverse makes our Big-Bang vantage a typical one among uncountably many (Chapter 4). And the framework's central claim is not "the universe might be like this" but the stronger, humbler "*from what we have already observed, something like us must exist*." We are not arguing the world into a shape; we are reading off what the measured world already entails. The work of the rest of this book is to make that entailment precise, quantitative, and falsifiable—and, where a more likely mechanism might exist that we have not yet found, to say so honestly. If a better one is found, it wins; until then, this is enough to require our own existence.

CHAPTER 2

No Singularities in Nature

A singularity is not a feature of the world; it is the noise a theory makes when it has been pushed past what it knows. General relativity, left to itself, predicts that gravitational collapse ends in a point of infinite density—and then it stops being able to say anything at all. The founding wager of this book (Chapter 1) is to take that breakdown at face value: physics is continuous and conservative, so the infinity is an artifact, and *something* must intervene before it is reached. In Einstein–Cartan gravity, something does. The collapse does not terminate. It *bounces*. Everything else in this book is the working-out of that single refusal to admit a singularity.

2.1 What stops a collapse

Einstein–Cartan gravity is general relativity with one restriction lifted: the connection is allowed an antisymmetric part, torsion, which couples to the spin of fermions (Axiom A3). At everyday densities this changes nothing measurable. But it adds a second term to the high-density stress–energy, and the two terms scale differently. The *gravitational* term is attractive and grows linearly with mass–energy density,

$$T_{\text{grav}} \sim \rho c^2,$$

while the *torsion* term is repulsive and grows as the *square* of the spin density s,

$$T_{\text{tors}} \sim \frac{G\hbar^2 s^2}{c^4}.$$

At nuclear density the ratio $T_{\text{tors}}/T_{\text{grav}}$ is about 10^{-40}—utterly negligible, which is why no laboratory has ever seen it. But compression packs more fermions into each volume, each carrying spin $\hbar/2$, and the quadratic term climbs faster than the linear one. Squeeze hard enough and repulsion overtakes attraction. The inward fall is halted from within, by the spin of the matter itself, and reverses. No outside agent, no new field, no fine-tuning: just a term that general relativity left out because torsion was assumed to vanish.

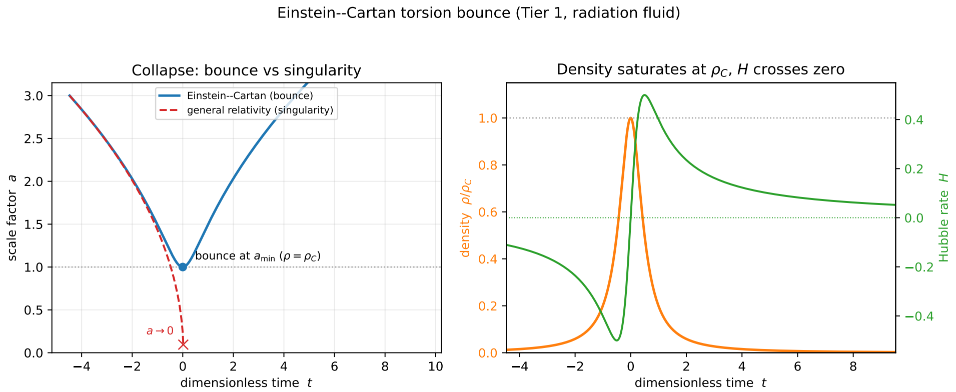

Figure 2.1: The Einstein–Cartan torsion bounce (a 1D spherical collapse, radiation fluid, in dimensionless units $8\pi G/3 = \rho_C = 1$). *Left:* the collapsing universe reaches a minimum radius $a_{\min}$ and re-expands (blue) exactly where general relativity would run to a singularity, $a \to 0$ (red dashed). *Right:* the density saturates at the bounce density ρ_C and the Hubble rate crosses zero—collapse turning into expansion, with no singularity ever reached. Reproducible as `figures/scripts/ch09_bounce.py`; the full numerical treatment is Chapter 8.

2.2 The bounce density

Setting the two terms equal fixes the density at which the reversal happens. For ordinary fermionic matter, with about one unit of spin $\hbar$ per nucleon mass,

$$\rho_C \sim \frac{m_n^2 c^4}{G\hbar^4} \sim 10^{50}\,\mathrm{kg\,m^{-3}}.$$

This sits 33 orders of magnitude above nuclear density and 46 below the Planck density—comfortably in a regime where Einstein–Cartan is still classical and trustworthy, far from where quantum gravity proper would be needed. The exact figure depends on the fermion content and on coefficients in the Einstein–Cartan Lagrangian, and pinning it precisely is one of the calculation targets of the simulation program (Chapter 8).

A name, and a pun. This is the one place the book indulges its title for the mechanism. The density at which Cartan's torsion forces the turnaround is where everything is dissolved and then extruded clean—a kind of *katharsis*. Cartan + *katharsis* gives *Cartasis*, and one may call ρ_C the Cartasis density and the bounce surface the Cartasis membrane. The pun is spent here; in the rest of the book they are simply the *bounce density* and the *bounce membrane*.

What is actually there. At ρ_C the familiar particle picture is gone. The matter is compressed past quark deconfinement, past electroweak restoration, past any regime in which the Standard Model's quasiparticles are even defined. What remains is continuous field content—energy–momentum density, spin density, chiral current density—and what crosses the membrane is therefore not *particles* but *conserved currents*: total energy–momentum, total spin, total chiral charge, and (subject to electroweak anomaly effects) baryon and lepton number. Nothing structured survives the crossing. Whatever a world is built from on the far side must be rebuilt from elementary field content—a point that will matter both for what a universe inherits (Chapter 4) and for why a stray ordered object cannot simply pass through (Chapter 3).

2.3 The geometry of the bounce

2.3.1 Radius becomes time

Inside a Schwarzschild horizon the radial coordinate r is timelike: future-directed motion is motion inward, and this is ordinary general relativity. The bounce uses that fact rather than fighting it. Torsion pushes the matter "outward" in r—but since r is timelike there, outward in r is not a path back through the horizon; it is a path into the *future* of the bounce. That future, read in the new region's own coordinates, is the cosmological future of a fresh spacetime. The parent's radial direction r becomes, smoothly across the membrane, the new world's time direction τ.

2.3.2 Continuity across the membrane

Because nothing is allowed to be discontinuous, the metric and connection must match across the membrane. The induced metric agrees from both sides; the extrinsic curvature matches under junction conditions of Israel–Darmois type, modified to carry torsion; and stress–energy and spin current flow through unbroken. These conditions are simply the precise statement of "the bounce is continuous"—a system of equations relating the parent-side and child-side fields at the surface, and the thing a faithful simulation must reproduce.

2.3.3 A persistent interface, not a single instant

A black hole that keeps accreting keeps feeding the bounce. New matter falls in, compresses, reaches the bounce density, and joins the membrane—so the surface is not a one-time event but a persistent interface that lasts as long as the parent hole does. From inside the new universe, this means matter is injected across the membrane throughout the parent's life, not only at the first instant. That ongoing injection is exactly what later supplies the dark sector (Chapter 5).

2.3.4 The parent's horizon is the child's first light

The same surface wears three faces. From the parent's exterior it sits just inside the event horizon. From the parent's interior it is wherever infalling matter is currently reaching the bounce density. From the child's interior, looking as far back in time as it can, it is the boundary of the past at maximum lookback—the child's Big-Bang surface. The parent's horizon region and the child's earliest sky are one surface seen from opposite sides. This single identification is what later lets the cosmic microwave background be read as the parent's Hawking glow (Chapter 5).

2.4 What the bounce does *not* do

The bounce reliably turns a black hole into an expanding world, but it does not also *make* the matter that fills it—it is no mirror that turns infalling antimatter into matter and hands us the cosmic excess for free. The interaction that drives it forbids that, and the proof is short. When torsion is integrated out of Einstein–Cartan–Sciama–Kibble gravity, the Dirac sector is left with a single four-fermion contact term, the Hehl–Datta interaction [8],

$$\Delta\mathcal{L}_{\text{tors}} = -\frac{3}{16}\,\frac{8\pi G}{c^4}\,(\bar{\psi}\gamma^5\gamma_a\psi)\,(\bar{\psi}\gamma^5\gamma^a\psi),$$

the negative square of the axial current. This is the very term whose energy density grows as the square of the spin density and overwhelms gravity at ρ_C: it *is* the bounce, so the bounce inherits its discrete symmetries. The axial current is even under charge conjugation, its contraction is even under parity, and the squared bilinear is even under time reversal. The interaction that drives the bounce is thus separately C-, P-, and T-even, and three things follow that no amount of modelling can wish away:

1. **No flip.** A C-even interaction cannot turn matter into antimatter. The membrane does not charge-conjugate what passes through it.
2. **No filter from nothing.** A C- and P-even interaction cannot *create* a chiral imbalance: zero net axial density in gives zero out.
3. **No arrow reversal.** A T-even dynamics does not run proper time backwards.

The bounce is real, robust, and generous—it reliably turns a black hole into an expanding world—but it is not a baryogenesis engine. Any matter–antimatter asymmetry must be *supplied*, not manufactured at the membrane. Where it actually comes from is the business of Chapter 4.

2.5 The arrow is struck at the bounce: a Janus point

If the bounce does not reverse time, where does a universe's arrow of time come from? It is *made* at the membrane. Along the matter's worldlines proper time runs monotonically; nothing turns around, the scale factor merely passes through a minimum. What turns around is the *thermodynamic* arrow. Infalling parent material arrives from a structured, high-entropy universe; compression toward ρ_C dismantles that structure and smooths the geometry—the Weyl curvature is driven toward zero—so gravitational entropy *falls* toward the membrane. On the far side, expansion lets structure, and eventually black holes, form again, so entropy *rises* away from it. Entropy is therefore at a *minimum at the membrane*, and the locally defined arrow points away from the bounce on both sides at once: a *Janus point* [1], one spacetime with a single monotonic proper time and two thermodynamic arrows diverging from a central low.

A newborn universe thus inherits a genuine low-entropy past for free, with no time flip required—the long-postulated "Past Hypothesis" arriving as a consequence of bounce geometry rather than as an unexplained initial condition. (That this low-entropy past is *forced*, and what it means for the timeless void from which the very first bounce is struck, is taken up in Chapter 3; how the arrow ends up locked to the matter–antimatter choice is taken up in Chapter 4.) The old slogan that "r becomes timelike, so time inverts" is a statement about Schwarzschild coordinates inside a horizon, not a physical reversal of anyone's clock.

2.6 What forms after the bounce

On the far side of the membrane the bounce dynamics open a new spacetime region—a child universe with its own time direction and its own spatial extent—and from inside it looks like a perfectly ordinary hot Big Bang: it begins at maximum density, expands fast, cools, passes through radiation- and matter-dominated eras, and grows large-scale structure. The early, very rapid expansion is driven by the torsion repulsion at its post-bounce peak and fades as the density drops, giving an inflation-like epoch with no separate inflaton; its quantitative shape, and the perturbation spectrum it should leave, are developed in Chapter 4 and put to the test in Chapter 6.

From the parent's side none of this is visible as a spatial object; the parent simply sees a black hole where the collapse occurred. The child's vast interior lies in the future of the bounce, in directions orthogonal to the parent's geometry—which produces the framework's most cheerfully counterintuitive feature, a *Tardis*: a universe far larger inside than its parent's horizon would suggest. Our own matter content, some 10^{53} kg, sits at the bounce density in about 10^3 m^3—less than half an Olympic pool—yet from inside we measure a universe tens of billions of light-years across. There is no contradiction: the small volume is the matter at the bounce moment in the parent's coordinates, the large one is the cosmological extent after expansion in the child's, and they are simply different quantities. Even plain general relativity already has the spatial volume inside a horizon growing without bound [4]; the bounce reinterprets that growth as the expansion of a child universe rather than the approach to a singularity we have agreed nature does not permit.

CHAPTER 3

The Void

The bounce of Chapter 2 turns a collapse into a world. But a bounce needs something to bounce—a region that reaches the bounce density in the first place. Step back, then, from any single universe to the largest picture the framework admits: the *void*, the eternal vacuum in which everything else is a rare condensation. This chapter asks what the void is, why it gives rise to anything at all, and—once it does—why what looks up at the sky is a real inhabitant of a real world rather than a fluke of order. The answers are, in order: a timeless maximum-entropy state; because emptiness can decay but cannot be improved upon; and because the void can give birth to a world but cannot dream up a mind.

3.1 What the void is

The supraverse is a single four-dimensional manifold—three space, one time, Lorentzian—of non-trivial topology. Almost all of it is vacuum: quantum-mechanically alive but unstructured. Embedded in that vacuum are condensed regions, the universes, each a connected sub-manifold with its own internal cosmology, joined to the rest through the bounce membranes of Chapter 2. The picture is foam-like—bubbles nested in parent–child relations—but, as the instanton calculation below will show, a very *dilute* foam: the bubbles are astronomically far apart, and emptiness is overwhelmingly the rule.

No fifth dimension is needed to hold this together. The intuition that one must "embed" a child inside a parent in some higher space is a visualization crutch, not a physical requirement: two regions joined at a bounce membrane is a perfectly good 4D manifold with a boundary identification, defined intrinsically. One *may* embed it in higher dimensions to draw it (Whitney guarantees one can), but the embedding carries no physics.

3.2 The void has no arrow

Before asking what the void *does*, we must be honest about what it is not: it is not a stage on which time runs. The microscopic laws—quantum field theory, general relativity, Einstein–Cartan—are all time-symmetric up to the tiny CP phase. An arrow of time is never in the laws; it is a *boundary condition*, appearing only where a system is handed a low-entropy edge. The supraverse as a whole has no such edge and needs none. To posit a global cosmic "now," some master clock for the void, would be to add unexplained structure; declining to costs nothing and is the honest, minimal choice.

So the right question about the void is not "which way does its time run?" but the timeless, thermodynamic one: *what is the stable state of the void?* That is an equilibrium question—an eigenstate question. Time is what clocks measure, and an equilibrium void has no clocks. Every arrow we will ever speak of is manufactured *locally*, at a bounce (Chapter 2), pointing away from that low-entropy membrane; time is emergent, created per universe, never inherited from the void. There is no contradiction between a timeless void and the vivid passage of time we feel—we feel it precisely because we live *inside* one of the void's condensations, on the high-entropy side of its bounce.

This dissolves an apparent paradox before it can take hold. The language of this book cannot help speaking of the void as though things happen in it—universes "nucleate," equilibrium is "established," a foam "coarsens." Those are descriptions borrowed from inside a condensation's own time and projected onto the whole. The void does not evolve *toward* equilibrium in some outer time; the equilibrium *is* the timeless stationary state, and "nucleation" is just the statement that this state already *contains* its rare condensations as part of what it is. No global clock ticks. Read every "then" that follows as the only tense we have, not as a claim about the void.

3.3 Why a void condenses

Naively, making universes looks entropically forbidden: ordered worlds out of formless vacuum. The naive view forgets gravity. Gravitational entropy runs *backwards* from the ordinary kind [2, 12]: a uniform matter distribution is low entropy (smooth metric, no horizons, nothing stored geometrically); clumped matter is higher; and a black hole is the maximum-entropy state for its mass–energy, $S = A/4\ell_P^2$, able to carry more entropy than everything that fell into it. So structure—clumps, horizons, black holes—is entropically *favoured*, not forbidden, and the Second Law does not oppose the appearance of worlds.

That much says condensation is allowed. It does not yet say how much of it there is, and here the careful answer overturns the naive one.

3.4 The maximum-entropy eigenstate

State "the stable state of the void" as the variational problem it really is, and the answer is sharper—and at first sight stranger—than "a foam." Take an infinite vacuum with a positive cosmological constant. On its own, such a vacuum is *de Sitter space*: it accelerates apart, and every observer is surrounded by a cosmological horizon at radius c/H_Λ, glowing with the Gibbons–Hawking temperature

$$T_{\mathrm{dS}} = \frac{\hbar H_\Lambda}{2\pi k_B} \approx 2 \times 10^{-30}\,\mathrm{K}$$

for our own Λ, and carrying the horizon entropy

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{cosmo}}}{4\ell_P^2} = \frac{\pi c^5}{\hbar G H_\Lambda^2} \approx 3 \times 10^{122}\, k_B. \quad (3.1)$$

Now ask what *maximises* the total horizon entropy of such a vacuum. In Schwarzschild–de Sitter space the generalised entropy is the sum of the black-hole and cosmological horizon areas, $S = (A_{\mathrm{bh}} + A_{\mathrm{cosmo}})/4\ell_P^2$, and inserting any mass shrinks the cosmological horizon faster than it grows a black-hole horizon. The sum is therefore largest at *zero* mass: **empty de Sitter space is the maximum-entropy configuration**—the stable state, the eigenstate.

This does not contradict the clumping argument of the previous section; it refines it. Clumping governs how an overdensity that is *already present* evolves—it is why a fluctuation that has reached the bounce density proceeds to bounce rather than disperse. But the global maximum, the eigenstate of the whole void, is the empty vacuum. Structure is locally entropy-increasing once seeded, yet globally entropy-*costly* to seed. The two readings meet in a single verdict: the void is overwhelmingly empty, dusted with the sparsest possible scattering of worlds.

Worlds are the eigenstate's faint components, not events in it. It is worth being exact about the word "fluctuation," because the void has no time. A genuine eigenstate is stationary: nothing in it *happens*, in sequence, ever. So the worlds are not a stream of events the void undergoes; they are the *timeless structure* of the one equilibrium state—the amplitude of that state having support, exponentially small, on configurations that already contain condensed universes. In this sense the void is a *perturbed* maximum-entropy eigenfunction: empty de Sitter is the dominant component, the worlds its subdominant ones, and the "perturbation" is not an external poke (which would need time) but the intrinsic nonlinearity of self-gravity, which opens a tunnelling channel out of the empty state. That a quantum state of pure gravity should be assumed at all—in this timeless, Wheeler–DeWitt sense—is the content of Axiom A1, and it is the conservative assumption: it adds no new law, only the universality of the law we have.

3.5 Why the seed of a world must be low-entropy

One might worry that we have only pushed the puzzle back: granted the void condenses, why is each new world born in the pristine, low-entropy state its forward arrow requires (Chapter 2)? The answer is that the seed *cannot* be otherwise. A condensation reaches the bounce density only by crushing its contents past every scale at which complex structure—and therefore information, and therefore entropy—can exist. The membrane is forced to be a state of high concentration and low complexity: uniform matter at the bounce density, with the Weyl curvature driven toward zero. That is exactly Penrose's low-gravitational-entropy initial condition (his vanishing-Weyl hypothesis [12]), and here it is not postulated but *forced* by the physics of the bounce. The low-entropy past that every universe needs is thus supplied for free, and the apparent regress closes: you cannot make a high-entropy bounce, because high entropy requires structure and structure cannot survive the crossing.

3.6 How the void makes a world

How often does the empty eigenstate decay into a seed? Because such seeds nucleate in the heat-dead de Sitter void of an older universe (the recursive cycle of Chapter 4), the birth rate is a de Sitter nucleation rate, and its scale is *set*, not free, by the void's dark energy. Per de Sitter horizon four-volume,

$$\beta = \lambda \frac{H_\Lambda^4}{c^3}, \qquad H_\Lambda \simeq 1.8 \times 10^{-18}\,\mathrm{s}^{-1}, \tag{3.2}$$

so the prefactor $H_\Lambda^4/c^3 \approx 4 \times 10^{-97}\,\mathrm{m}^{-3}\mathrm{s}^{-1}$ is fixed by the observed dark energy—about one nucleation *attempt* per Hubble four-volume. The whole question is the dimensionless suppression $\lambda = e^{-I}$, with I the action of the gravitational instanton that carries the empty vacuum across the bounce density.

That action has a closed form. Nucleating a mass M in de Sitter costs the Boltzmann weight $e^{-Mc^2/k_B T_{\mathrm{dS}}}$, and that exponent is exactly the cosmological-horizon entropy deficit of inserting M, so

$$I = \frac{2\pi M_{\mathrm{seed}}\, c^2}{\hbar H_\Lambda}. \tag{3.3}$$

The only input is the lightest seed that can actually bounce: a blob at the bounce density turns into a universe only if it is self-gravitating ($R_s \simeq R$), which fixes $M_{\mathrm{seed}} \sim 9 \times 10^{14}$ kg—about the mass of a mountain. A lighter blob has no horizon and simply disperses. Putting that in (Figure 3.1),

$$I \;\sim\; 2.5 \times 10^{84} \;\gg\; I_{\mathrm{crit}} \simeq 1.4, \tag{3.4}$$

where I_{crit} is the de Sitter percolation threshold [7] that separates a void which tiles into a packed foam ($I < I_{\mathrm{crit}}$) from one that inflates faster than it nucleates, leaving worlds isolated ($I > I_{\mathrm{crit}}$). The margin is eighty-four orders of magnitude; no semiclassical correction, no factor-of-ten ambiguity in the seed mass, can close it. So $\lambda = e^{-I}$ is, for every practical purpose, zero, and the verdict is unambiguous: **the supraverse is dilute**—a sparse, ever-inflating scatter of isolated, round island universes, each a single bounce, astronomically far from its neighbours. This lands the framework squarely in the eternal-inflation family, and it is the picture the rest of the book assumes; what those island universes are like, how they grow, and how they seed the next generation is the subject of Chapter 4.

3.7 Why we are not Boltzmann brains

The dilute verdict revives the oldest objection to any fluctuation cosmology—and disposes of the suspicion that we are a simulation. If a whole universe is

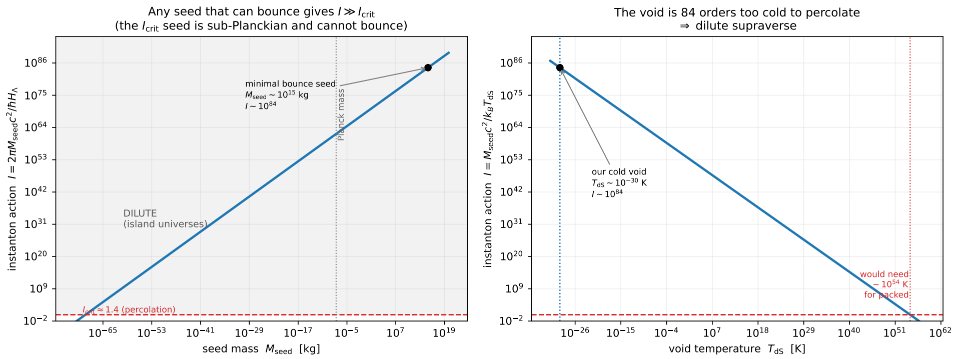

Figure 3.1: The OG-nucleation instanton action $I = 2\pi M_{\rm seed}c^2/\hbar H_\Lambda$ (Equation (3.3)). *Left:* I versus seed mass; any seed massive enough to bounce ($\sim 10^{15}$ kg, the self-gravitating minimum) gives $I \sim 10^{84} \gg I_{\rm crit} \simeq 1.4$. *Right:* I versus void temperature; the cold de Sitter void ($\sim 10^{-30}$ K) is eighty-four orders too cold to percolate. The verdict is robust: a dilute, eternal-inflation-like supraverse of isolated island universes. Reproducible as `figures/scripts/ch05_instanton.py`.

this fantastically rare, surely a single mind, with its memories, is cheaper to fluctuate into being than an entire cosmos; and then a *typical* observer should be such a fluctuation—a "Boltzmann brain"—not a real inhabitant of a real world. Cast in the de Sitter Boltzmann weights above, the bare costs seem to agree:

$$\underbrace{10^{-10^{69}}}_{\text{a brain }(\sim 1\,\text{kg})} \gg \underbrace{10^{-10^{84}}}_{\text{a world-seed }(\sim 10^{15}\,\text{kg})} \gg \underbrace{10^{-10^{123}}}_{\substack{\text{a finished low-entropy}\\ \text{Big Bang [13]}}}, \tag{3.5}$$

the brain winning by the unbridgeable factor $10^{10^{84}}$. And no rescue helps—not the head-count of real observers a universe produces, not the recursive tree of its descendants, not a real life's length against a brain's single flickering thought. Each of those is a *single* power tower, a number of at most a few thousand digits, and none can dent a *double* tower of height 10^{84}.

The resolution is not to win the count; it is to see that it is the wrong count, because *the brain and the bare Big Bang cannot come from the void at all.* A Boltzmann brain—or, what is the same thing, the optimal computer running a simulation of one—is an information processor, and §3.2 fixed what such a thing requires: metabolism, memory, and computation are all dissipative, and dissipation needs an arrow and a low-entropy reservoir to discharge into. The void offers neither. It is the stationary, arrowless, maximum-entropy eigenstate of §3.4—with no dynamics to fluctuate and no gradient to compute against. The brain is parasitic on an arrow it cannot itself create; the $10^{-10^{69}}$ was reckoned as though the void were a churning thermal bath, which it is not [3].

What the eigenstate *can* do is decay. Nucleation from a stationary state is a genuine quantum transition—the gravitational instanton of §3.6— and it yields only the *generic* state of the mass-energy it creates: a structureless, maximal-entropy, black-hole-like blob. That blob is exactly a viable seed; it carries no improbable order of its own, yet when it bounces, the bounce is *forced* to be low-entropy (§3.5), manufacturing order and arrow for free on the far side. A brain is not a blob. It is a low-entropy *organised* state, and nucleation cannot deliver organisation—only generic lumps—while the arrowless void cannot assemble one afterward. So the void makes seeds, never brains.

The same blade cuts Penrose's bare Big Bang, and this is the deep part. A hot Big Bang handed over whole in its 1-in-$10^{10^{123}}$ low-entropy starting state [13] is *also* an ordered, arrow-bearing configuration posited with no mechanism to make it. It presupposes the very low-entropy past it was meant to explain, so it too cannot be nucleated from the void—it can only be put in by hand. A cosmology whose sole origin story is "the initial state simply happened to be that special" is, structurally, the simulation hypothesis: it declares us an un-generated fluke of order with no maker. *If the bare Big Bang were the only available beginning, we would indeed be a simulation.* The framework escapes because it never posits the order: it pays $\sim 10^{-10^{84}}$ to nucleate a generic seed and lets the bounce force the 10^{123} worth of low-entropy order at no extra cost—turning Penrose's improbability-of-order from an unexplained initial condition into a consequence of geometry at the membrane.

Two levels then close the problem. At the *foundation*, nothing mind-like can bottom out the stack; only an arrow-creating bounce can, so the base layer of reality is real worlds, not a sea of fluctuations. *Inside* a world the arrow exists, so brains *can* fluctuate—but only over the de Sitter recurrence time $\sim 10^{10^{69}}$ years, and our dark energy is finite (it is the parent's accretion, Chapter 5, and switches off when the parent finishes evaporating, $\sim 10^{136}$ years), recycling our matter into new bounces long before a single interior brain could form. A ΛCDM model with a truly eternal cosmological constant has no such escape—its endless de Sitter future is a literal Boltzmann-brain factory—so this is a respect in which the framework is *strictly better behaved* than the standard model, not merely different.

Epistemic status. The de Sitter measure problem—whether and how a stationary vacuum "fluctuates"—is not settled, and a fully rigorous measure for the framework does not yet exist (it is among the open questions of the appendix). What is robust is the qualitative asymmetry: nucleation yields generic blobs, blobs bounce into arrowed worlds, and organised low-entropy minds presuppose an arrow they cannot source from an arrowless eigenstate. The cognitive-stability argument seals it from the other side—any theory predicting that typical observers are fluctuations dissolves its own evidential basis, since a fluctuated mind has no warrant to trust the memories and reasoning that built the theory, including the three numbers in (3.5). We are entitled to conclude we are real not by counting, but because the alternative refutes itself and because the void's only product is the real thing. Reproducible as `sims/src/cartasis_sims/boltzmann_brains.py` and `void_eigenstate.py`.

CHAPTER 4

Eternal Dawn

The void decays into seeds (Chapter 3), and each seed bounces (Chapter 2) into a world. A world born this way—directly from the void, with no parent outside it—is an *original-generation universe*, an OGU: the first dawn of a lineage. This chapter asks what those first universes are like—what they inherit, how they choose between matter and antimatter, what sizes and spins they come in—and how, by growing old and dying, each one seeds the next, so that the dawn is not a single event but an eternal, recurring one.

4.1 The first dawn, and what a universe inherits

Just past the bounce the new universe is at its densest, and the torsion repulsion that halted the collapse is still near its peak. It drives a brief epoch of very rapid expansion that fades as the density drops and gravity reasserts itself—an inflation-like phase with no separate inflaton field, ending on its own when the density falls below the bounce threshold, and leaving behind the homogeneous, near-isotropic conditions a hot cosmology needs. Deriving its perturbation spectrum and matching it to the microwave sky is the central near-term test of the whole picture (Chapter 6).

What actually comes through the membrane is set by conservation laws, not by inheritance of any structure. Energy–momentum is conserved, fixing the new universe's mass budget. Net angular momentum passes through, so a rotating seed or parent leaves the child with a *preferred axis*—a candidate source of the large-scale anisotropies discussed later. The conserved charges flow through, in particular the combination $B - L$ that will carry the matter excess. What does *not* come through is everything made of structure: atoms, nuclei, particle identities, any record of a "before." All of it is dissolved at the bounce density and rebuilt from elementary field content on the far side (Chapter 2). A universe inherits a budget and a few conserved numbers, not a cargo.

4.2 The matter choice

One of those conserved numbers is the most consequential the universe will carry: the slight excess of matter over antimatter, measured today as the baryon-to-photon ratio $\eta \simeq 6 \times 10^{-10}$, about 0.6 parts per billion. Chapter 2 established that the bounce cannot *make* this excess—the interaction that drives it is C-, P-, and T-even, so it neither flips charge nor generates a chiral bias from nothing. The asymmetry must be *supplied*. The framework's account of where it comes from divides cleanly by generation.

For a child universe: inheritance, essentially exact. A universe with a parent inherits the parent's excess directly. At the bounce the infalling material thermalises at $\sim 10^{20}$ K into a near-symmetric bath in which every species is pair-produced; the parent's particular matter is erased. What survives is whatever net $B - L$ the material carried, because the electroweak sphalerons active at the bounce temperature wash out $B + L$ but cannot touch $B - L$. The crossing is also fast—$\sim 4 \times 10^{-21}$ s, far shorter than any thermalisation time—so it generates essentially no entropy (the dilution factor $D = S_{\text{after}}/S_{\text{before}} = 1$ to about one part in 10^{11}; Chapter 8). With entropy conserved, the child does not regenerate its photons, and so it inherits the parent's ratio outright, $\eta_{\text{child}} \approx \eta_{\text{parent}}$. The excess is passed down a lineage like a family name, not re-manufactured at each step.

For the first universe: the seed's lucky skew. Only the OGU, with no parent, must source its excess at birth—and even there nothing needs to *make* it. A vacuum fluctuation rare enough to nucleate a bounce is not exactly matter–antimatter symmetric; it carries some chance net $B - L$. The symmetric part incinerates in the thermal bath, and the surviving skew fixes that lineage's η. A quantitative attempt to source the skew dynamically—the chiral-vortical effect, in which the rotating bounce's vorticity ω drives an axial current and freezes in a bias $\eta \sim C\,(\omega/T)$—gives the right *order* but not, yet, the right number: at the bounce density the bare, parameter-free ratio is $\omega/T \sim 1.5 \times 10^{-11}$, landing within about 1.6 orders of magnitude of η_{obs} when it could as easily have been 10^{-30}, but matching the value would require an uncomputed coefficient $C \cdot f_\omega \simeq 41$. This is a scaling estimate with explicit assumptions, not a derivation (reproducible as `sims/src/cartasis_sims/cve_filter.py`), and it binds *only* the OGU; for the rest of us the excess is inherited debris, and the shortfall is not our problem.

The needle, threaded by selection. The surviving excess must thread a fine target: too much and the universe is over-baryonic, too little and it annihilates to a sterile furnace; the viable window sits at about one part per billion. The framework does not fine-tune to hit it—it *selects*. Only lineages founded on a sufficiently skewed seed are matter-rich enough to make stars, black holes, and descendants at all (§4.3); the barren ones make nothing and are never anyone's parent. We carry down what a viable founder happened to have. What remains genuinely open is not the stability of η along a lineage but its *value* at the founder—why $\sim 10^{-9}$ rather than 10^{-3} or 10^{-15} (see the appendix).

Why matter comes paired with a forward arrow. The thermodynamic arrow of Chapter 2 is set by the low-entropy membrane and would exist even in a perfectly symmetric universe; it is not *caused* by the matter excess. But the two are co-sourced at the same surface and locked together across the supraverse. The process that biases the matter is CP-violating, and by the CPT theorem CP violation is T violation, so matter genesis is intrinsically time-asymmetric. Because the void is CPT-symmetric, for every matter-led universe running forward there is a CPT-image partner that is antimatter-led and runs the opposite way—the same object seen through CPT. The quantity fixed by symmetry is the *product* of matter-sign and arrow-sign, and observership selects the diagonal: every observer necessarily finds "matter, forward in time," while the inhabitants of the mirror lineage say exactly the same of their "antimatter, backward" world. Matter and the arrow travel together not because one causes the other, but because the CPT structure of the void binds their signs.

4.3 The shape of a generation

A generation of first universes is not uniform; it has a derivable distribution in both size and spin, and we should expect to find ourselves at its typical point, not in its rare tail.

Size: small is typical. Two pressures compete. Births favour the small—a fluctuation is exponentially less likely to reach the bounce density over a larger mass. Growth favours the large—once born, a hole feeds by runaway accretion. In the stationary population these balance into a power law, $n(M) \propto M^{-g}$ (with $g \simeq 1$–2 for Eddington- to Bondi-limited feeding): many small universes, few large, cut off at the high-mass end (Figure 4.1, left). Since the count falls with mass, a low-mass viability cut—enough mass to make stars and black holes—selects observer-bearing universes that pile up just above the edge. The old conjecture that "we sit near the optimum" is thus *derived*: a typical observer is near the viability edge of a declining power law, not out in the exponentially rare massive tail.

Spin: barely viable is typical, and that is why η is small. The seed's net vorticity sets the inherited bias, $|\eta| \sim C\,|\omega/T|$. A random fluctuation most likely has *small* net vorticity, but a seed below $\omega_{\min} = \eta_{\min} T/C$ cannot complete baryogenesis: its matter and antimatter annihilate to near-symmetric radiation, no structure forms, and the universe is sterile. Viability therefore gates the distribution from below, and the result runs exactly counter to intuition (Figure 4.1, right): *no spin is likeliest but most sterile, high spin is rare but pure*, and the typical *viable* universe sits just above the threshold—low-spin, low-purity, barely past barren. This is a shape, not yet a number (it is plotted in units of an unknown birth-vorticity scale), but it carries a real payoff: it explains why the cosmic matter asymmetry is *small without fine-tuning*. A small η is what a typical viable universe has. The same low spin predicts a small but nonzero net rotation for our universe—a preferred axis to confront with galaxy-spin handedness and CMB alignments (Chapter 6).

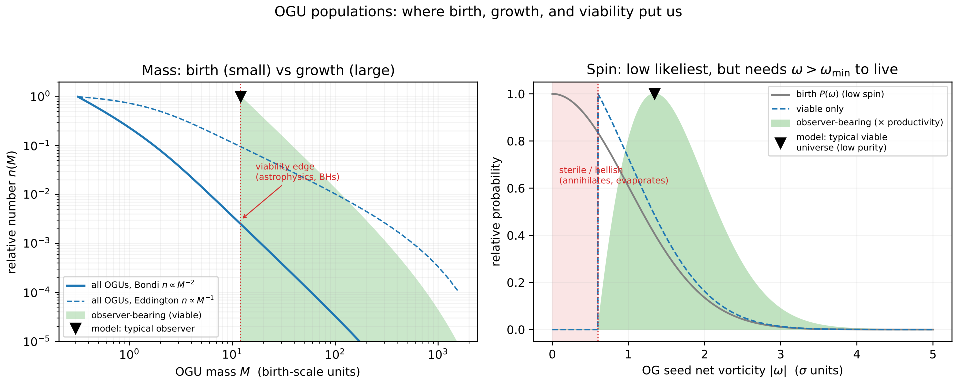

Figure 4.1: The two derivable population shapes. *Left:* the mass distribution $n(M) \propto M^{-g}$—births (small) versus runaway growth (large)—with a viability cut selecting observer-bearing universes, which peak just above the edge. *Right:* the spin distribution—low net vorticity is likeliest, but below $\omega_{\min}$ baryogenesis fails and the universe is sterile, so viable universes sit just above threshold (low purity, small η). The markers are the model's predicted typical location, not measurements of our universe; the axes are in model units. Generated by `figures/scripts/ch05_distributions.py`.

Lineages are chirally pure. Because the sign of the excess is set once, at the seed, and merely passed down thereafter, a matter-led OGU has only matter descendants and an antimatter-led one only antimatter, with no mixing. The vacuum's CPT symmetry makes the two kinds equally numerous, so global CPT is preserved even though every individual lineage is pure. What is inherited is the chirality *sign* alone—*not* the spin: a descendant's rotation axis and magnitude are its own progenitor hole's, set by a local astrophysical collapse and re-drawn every generation. Our spin is our progenitor hole's, not our distant founder's (Chapter 6).

4.4 Growth, death, and the eternal recurrence

A first universe, once viable, grows—its central holes feed and merge, its mass climbs—and it lives a very long time: the parent black hole that hosts a universe of our mass would take on the order of 10^{145} years to evaporate, far longer than any internal cosmological clock. But it is not immortal. Its stars burn out, its black holes themselves Hawking-evaporate, its matter redshifts away, and the interior settles into cold ($\sim 10^{-30}$ K), empty, dark-energy-driven de Sitter space. And a vast, empty, smooth de Sitter interior is—gravitationally—*a void*. It is exactly the substrate Chapter 3 began from. So it does what a void does: it nucleates a fresh, dilute scatter of new first universes inside itself, and the cycle turns.

The void is not primordial. This collapses a hidden assumption. The void of Chapter 3 is not a separate arena that existed "before" everything; it is the asymptotic state every universe reaches. Universes nest both ways—we are

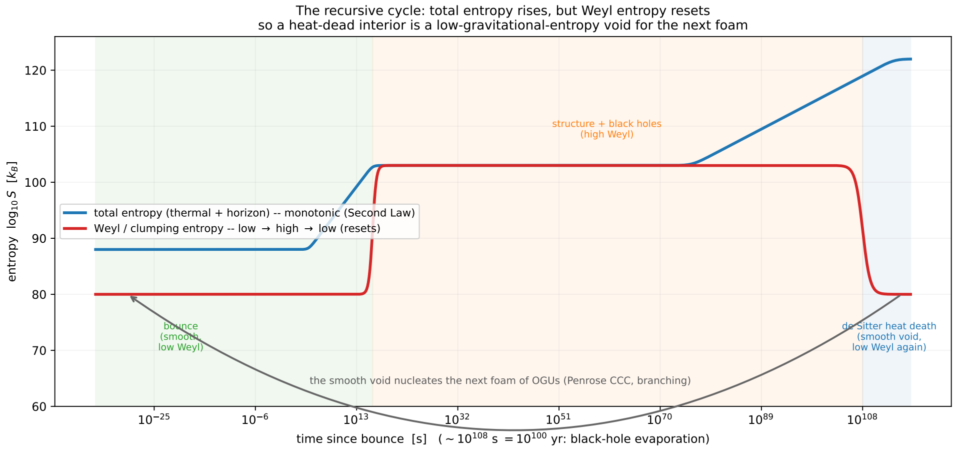

Figure 4.2: Why a heat-dead interior can seed the next foam. The *total* entropy (rising) climbs monotonically to the de Sitter maximum, but the *Weyl/clumping* entropy (the one a fresh bounce needs small) cycles—low at the smooth bounce, high while black holes exist, low again once they evaporate and the interior smooths to de Sitter. So the heat-dead void is a valid low-gravitational-entropy substrate for the next foam: Penrose's conformal cyclic cosmology, made to branch. Schematic, with order-of-magnitude entropies on the black-hole Hawking timescale.

inside a parent, and our own far future will, in its emptiness, seed a foam of new lineages. There is no first void and no last universe. The supraverse is self-similar and eternal in both directions, and asks for no first cause.

Why heat death does not forbid it. The obvious objection is that heat death is *maximal* entropy while a fresh bounce needs *low* entropy. Penrose's resolution applies exactly [13]. The *total* entropy rises monotonically, obeying the Second Law; but what a bounce needs small is the *gravitational* (Weyl, clumping) entropy, and that *cycles*—low at the smooth bounce, high while matter clumps into black holes, low again once the holes evaporate and the interior smooths back to conformally flat de Sitter (Figure 4.2). It is the Weyl entropy a new bounce needs low, and it resets every cycle. This is Penrose's conformal cyclic cosmology with one change: a dead aeon here does not hand off to a single successor, it nucleates *many*—a whole foam. The framework is conformal cyclic cosmology *made to branch*. (The recurrence is a synthesis at the speculative end of the program—a natural closure of the picture, consistent with the gravitational arrow, not a derived result. Reproducible as `figures/scripts/ch05_cycle.py`.)

And so to us. Inside any viable universe, stellar collapse makes black holes, and each of those is the seed of a child universe—a black-hole universe, a BHU, the next rung down the recursion. We are on one of those rungs, not at the top: our universe carries dark matter and dark energy, and in this framework both *require a parent* (they are the parent's mass and ongoing accretion felt through the membrane, Chapter 5). An OGU, born straight from the void, would have neither. So the very darkness of our sky tells us we are a child, not a firstborn—which is the subject of the next chapter. It also rescues a Copernican worry the framework had quietly incurred: to find ourselves a few billion years into a Big Bang looks special, until the recurrence reveals it as the most ordinary thing there is. Every universe has its own bounce and its own observers a few billion years on; there have been and will be endless aeons, each a foam of universes nested in the heat-dead past of another. We sit not at a privileged origin but at a typical vantage—one lit pixel in an eternal, self-similar screensaver of near-identical dawns.

CHAPTER 5

Home

We have followed the framework from the timeless void (Chapter 3) through the first dawn (Chapter 4) down the recursion to its children. This chapter arrives home—at our own universe—and finds that the parts of the sky cosmology has had to *add by hand*, the dark sector, are here not additions at all but the plain look and feel of living inside a black hole. Dark matter, dark energy, the microwave background, the great rings in the galaxy distribution: read in this framework, each is a feature of our address, and together they are a family portrait of the parent on the far side of our horizon.

5.1 We are a child

One observation settles our place in the family tree before any detailed accounting. Our universe contains dark matter and dark energy, and in this framework both *require a parent*: dark matter is the parent's mass felt through the persistent bounce membrane, and dark energy is the parent's ongoing accretion (this chapter makes both precise). An original-generation universe, nucleated straight from the void, has no parent and so would have neither. Their presence in our sky is therefore a direct reading:

$$\text{(observed dark matter and dark energy)} \implies \text{we are a child universe—a black-hole universe, generation } n$$

We are not a firstborn. The darkness of the sky is the evidence.

5.2 The persistent channel

The bounce membrane is not a one-time event but an ongoing interface (Chapter 2). As long as the parent black hole keeps accreting, matter keeps falling in, reaching the bounce density, and crossing into our interior—throughout the parent's entire life. But the injection geometry is peculiar, and everything in this chapter follows from it: new matter does not appear at some location in our three-dimensional space. It appears *at the membrane*, which from inside is our cosmic horizon at maximum lookback—our Big-Bang surface. Injected matter therefore manifests as gravitational influence arriving from the cosmic boundary, not as particles materialising among the galaxies.

> **Why you can seed a child but never visit one.** It is worth seeing what kind of surface this is, because it answers a natural question:

could one, granted every speculative tool relativity allows—warp drives, traversable wormholes, exotic matter—*travel* to the parent, or to a baby universe forming in our sky? No, in both directions, for the same reason. The membrane is not a wall in space, which a warp bubble could defeat, but a surface in *time*: a bounce, a low-entropy boundary. The parent lies behind our Big-Bang surface, in our *past*; you cannot drive to your own Big Bang. And a baby universe is a real black hole you *can* fall toward—but its time runs forward from its bounce, so the horizon delivers you to its low-entropy first instant, not to any interior "now." You could become a speck in its initial conditions; you can be an ancestor, never a tourist. The membrane-as-Big-Bang structure is the framework's own chronology protection: you may fall down the recursion, never climb back up.

5.3 Dark matter: the parent's gravity

The matter on the parent's side of the membrane has mass–energy, so it curves our space; but it does not enter our interior, exchange photons with our matter, or annihilate against it. That is precisely the phenomenology of dark matter: halo-like mass around galaxies, extra rotational binding, lensing without visible source, large-scale correlations—all gravitational, none luminous. In this framework it is not a new particle but the live gravitational signature of the parent's ongoing, biased accretion.

Gravity felt before light seen. Because the membrane is our *past* horizon, the parent's mass is already in our metric—it curves our space *now*—while its light is still climbing in toward us from that receding past surface, and will for cosmic ages. So the dark sector's gravitational pull *leads* its electromagnetic visibility: we feel matter we cannot yet see (Figure 5.1). "Dark" matter is, on this reading, exactly gravity arriving ahead of light from the membrane.

Perhaps the missing antimatter. If the bounce has any chirality-selective character, the rejected handedness remains at the membrane—gravitationally present, geometrically separated from our interior, unable to annihilate against our matter. Cosmology's "missing antimatter" and its "dark matter" may then be one phenomenon. The reading even carries a number: if a fraction f of the infalling matter passes the extrusion, the dark-to-baryonic ratio is about $(1-f)/f$, so the observed $\sim 5{:}1$ implies $f \sim 1/6$—a target a first-principles Einstein–Cartan calculation should hit, and a sharp way to be wrong (Chapter 8).

What distinguishes it from a particle. Geometric dark matter need not clump exactly as cold particles do, need not appear in any direct-detection experiment, and should trace the parent's projected mass rather than our own luminous structure—tests taken up in Part II (Chapter 6).

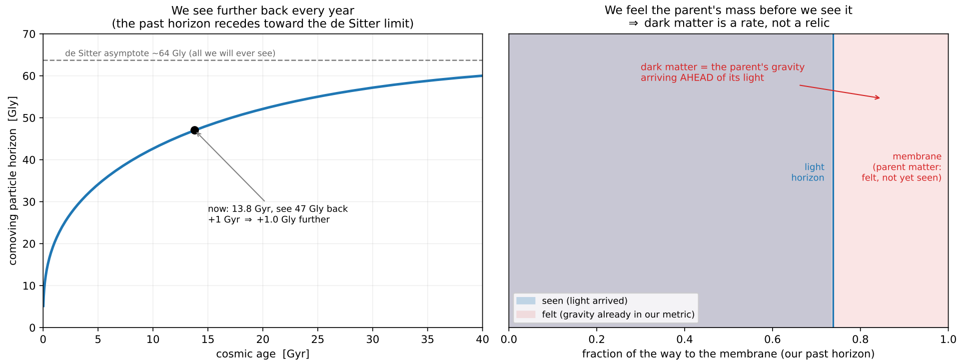

Figure 5.1: Gravity felt before light seen. The parent's matter sits at (and beyond) our past cosmological horizon—the membrane—so its mass is already in our metric and pulls on us now, while its light is still climbing in from that receding surface. The dark-matter gravitational signature therefore leads its electromagnetic visibility. Reproducible as `figures/scripts/ch06_lookback.py`.

5.4 Dark energy: the parent's growth

The parent is not a static reservoir; it is a growing black hole, feeding from its own universe. As its mass climbs, the membrane's geometry relative to our interior changes, and from inside that shows up as an extra outward push on our expansion—the acceleration that standard cosmology attributes to dark energy. This makes dark energy a *rate*: to leading order its strength tracks the *time-derivative* of the parent's accretion. A parent still gorging sustains or strengthens it; a parent whose surroundings are thinning, or which is turning toward Hawking-dominated mass loss, yields weakening dark energy—and, in the far future, a reversal, as the membrane shrinks and our expansion decelerates. So dark energy here is neither constant nor eternal: it is a readout of our parent's life story, and the recent hints from DESI that its equation of state may be evolving are exactly where to confront it (Chapter 6).

5.5 The microwave background: the parent's glow

The membrane wears, we have seen, three faces: the parent's event horizon from outside, our cosmic horizon at maximum lookback from inside, the Big-Bang surface in the standard telling. A black hole's horizon radiates—Hawking's result—and from our interior that radiation arrives from every direction, at maximum cosmic lookback, with a thermal spectrum. That is the cosmic microwave background. The parent's Hawking temperature at emission is essentially zero for so large a hole, but the radiation has been redshifting through our entire expansion; the 2.7 K we measure is what survives. The identification sits well with what the background *is*: a near-perfect Planck spectrum (Hawking radiation is thermal), near-perfect isotropy (the membrane is smooth across the parent's horizon), and anisotropies at the

10^{-5} level (membrane inhomogeneity and a little parent rotation). The one genuine debt is the acoustic peak structure, which standard recombination fits beautifully and which a bounce cosmology must re-derive—the central calculational challenge, taken up in Chapter 8.

If the parent rotates, its axis imprints a preferred direction on the membrane, and so on our earliest sky and the structure that grows from it. The reported large-angle CMB alignments—the "axis of evil"—and the JWST galaxy-spin handedness asymmetry are candidate signatures of that single inherited axis; the framework's sharp prediction is that they should *share* it (Chapter 6).

5.6 Lumpiness: the texture of the meal

What the parent feeds in is not smooth. A growing hole eats streams, clumps, the cosmic web of its own universe, so the matter projected through our membrane carries the parent's inhomogeneity—structure on scales unrelated to our own internal history. This sharpens into a structural signature of the recursive supraverse. Trace lumpiness down a lineage: the void is homogeneous; an OGU, with no parent, inherits none and (lacking dark matter) builds only weak internal structure—it is the minimal-lumpiness rung, and *this is exactly what* Λ*CDM assumes*, a smooth near-singular start with structure built only from within. In the language of this framework, ΛCDM describes an OGU. But a child inherits its parent's lumpiness *and* adds its own, so large-scale power *accumulates* generation by generation toward a fixed point, with the biggest jump at the first child, where a parent's lumpiness appears for the first time (Figure 5.2).

The observable falls straight out: our universe should show *more* ultra-large structure than any smooth-start model predicts. And it does—the Giant Arc, the $\sim$400 Mpc Big Ring [11], the Sloan Great Wall, the KBC void, all exceeding the $\sim$ 260 Mpc homogeneity scale and resisting assembly in the time available. In ΛCDM these are an embarrassment; here they are simply the inherited rung. Lumpiness thus joins dark matter and dark energy as a third, independent sign that we are a child and not a firstborn. The model is illustrative—the intrinsic and transferred contributions are not yet derived from the membrane projection—but its robust content is a direction a singular start cannot produce: monotonic excess large-scale power for every generation past the first.

None of this is bolted on. Dark matter, dark energy, the background glow, the great rings: each is a different glimpse of the same parent pressing through the same membrane, and together they place us—unspecially—inside one black hole among uncountably many. That is the framework's picture of home. Whether the picture is *true* is the question Part II takes up: what we can already see, what the next decade will decide, and what only computation can yet settle.

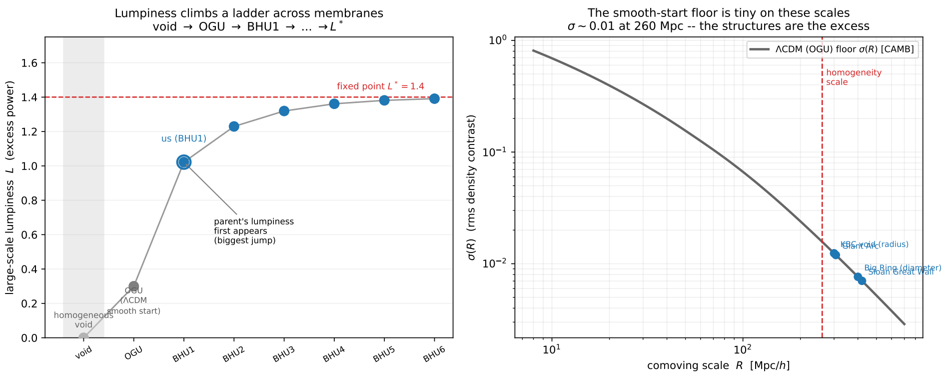

Figure 5.2: The lumpiness ladder. *Left:* large-scale lumpiness climbs from the homogeneous void through the OGU (minimal—the ΛCDM smooth-start case) to the first child (us; the biggest jump, where the parent's lumpiness first appears) and on toward a fixed point. *Right:* the smooth-start floor $\sigma(R)$, computed with CAMB, falls to ~ 0.01 by the 260 Mpc homogeneity scale, so the observed ultra-large structures (markers) sit far above what a singular start builds—the inherited rung that marks us a child. Reproducible as `figures/scripts/ch06_lumpiness.py`.

Part II

Observations and Simulations

CHAPTER 6

What We Can Already See

Part I built the framework; Part II asks whether it is true. The honest place to start is with what is *already* on the table—phenomena observable today, several of them long-standing puzzles for the standard model. None is yet a confirmation, and this chapter says so for each. But the pattern is suggestive: a single inherited axis, a dark energy that evolves, a Hubble tension that inhomogeneity resolves for free, structure that is too lumpy and too early, and a matter excess pinned twice over. Each is exactly what living inside a black hole (Chapter 5) would look like from the inside.

6.1 A preferred axis

If our universe inherited a rotation axis from its parent (Chapter 4), that single direction should show up in independent places: in the handedness of galaxy spins and in the large-angle pattern of the microwave background. Both hints exist. The JWST JADES survey reports a roughly 2 : 1 asymmetry in galaxy rotation directions, peaking near the Galactic poles, at 3.39σ [14]; and the CMB's low multipoles show the long-standing "axis of evil"—quadrupole and octupole aligned with each other and with the ecliptic. The framework's prediction is that these are two views of one axis, the parent's spin imprinted on our earliest sky.

The measurement is hard, and honesty demands showing why. Run the spin test on the largest clean catalogue available—the 30,126 Galaxy Zoo 1 spirals [10]—and the result is a cautionary tale (Figure 6.1). There is a highly significant *overall* handedness offset (a 3.9% clockwise deficit, $p \sim 10^{-11}$), but a label-shuffle null shows it is the known human perception bias [9], not a sky dipole ($p = 0.62$ beyond the monopole), and the best-fit axis lies 7° from the Galactic pole but 35° from the CMB axis. On the one large catalogue we have, the asymmetry is systematics-dominated and what axis exists tracks the Milky Way, not the cosmos. This does not refute the prediction—Galaxy Zoo 1 is a northern footprint with uncontrolled bias—but it fixes what a real test needs: an all-sky, mirror-validated catalogue.

The decisive test, and what it can and cannot do. Both confounded clues—the galaxy asymmetry near the Galactic pole and the CMB axis near $(\ell, b) \simeq (260^\circ, 60^\circ)$—reduce to one sharp question: does a clean galaxy-spin axis fall

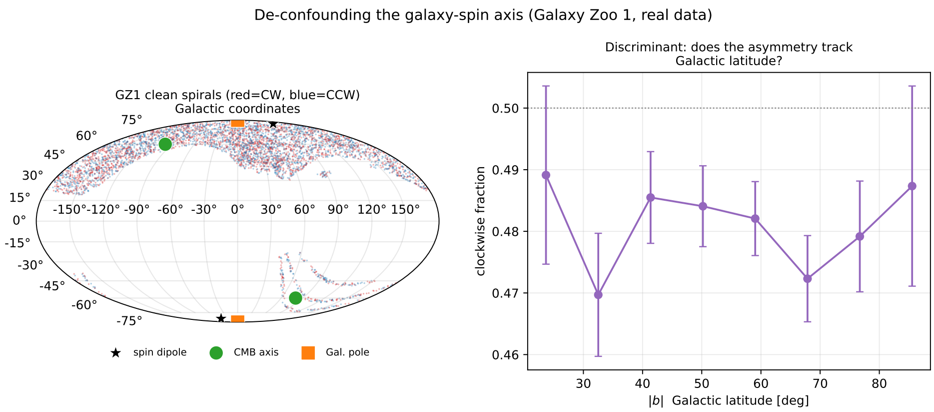

Figure 6.1: De-confounding the galaxy-spin axis on Galaxy Zoo 1 (clean spirals). *Left:* the SDSS northern footprint (red clockwise, blue counter); the fitted spin dipole (star) sits on the Galactic pole, 35° from the CMB axis. *Right:* the clockwise fraction sits a few percent below 0.5 independent of Galactic latitude—a global perception offset, not a significant dipole (shuffle null $p = 0.62$). Reproducible as `figures/scripts/ch08_galaxy_dipole.py`.

on the CMB axis (the parent imprint) or on the Galactic pole (a Milky-Way bias)? As a two-hypothesis comparison the log-likelihood ratio is

$$\ln \frac{L_{\mathrm{axis}}}{L_{\mathrm{sys}}} = \frac{\theta_{\mathrm{sys}}^2 - \theta_{\mathrm{axis}}^2}{2\sigma^2}, \tag{6.1}$$

with σ the precision to which the axis is measured. The two candidate poles are only $\simeq 30^\circ$ apart, so resolving them is purely a matter of precision: $\sigma \sim 30^\circ$ leaves them confused, while $\sigma \lesssim 10^\circ$ separates them at better than 100 : 1 (Figure 6.2). That is all the "30° versus 10°" means— the targets sit 30° apart, so the aim must be $\sim 10^\circ$. The test is deliberately *asymmetric in what it can conclude*. Because spin is local and most viable universes are low-spin (Chapter 4), a quiet sky is the *typical* expectation, not a failure:

- **Confirmation (strong):** a clean galaxy-spin axis that *agrees* with the CMB axis—one shared cosmic axis across independent probes, hard to fake.
- **Falsification (strong):** two clean, significant axes that *disagree*—a single parent spin cannot imprint two directions.
- **Inconclusive (the likely near-term case):** a null, or an axis on the Galactic pole—consistent with a low-spin parent *and* with systematics, so it neither confirms nor refutes.

So the shared axis is genuinely *confirmatory* and only weakly disconfirmatory, and it bears on this *one* prediction alone—the bounce, our child-universe placement, the CMB peaks, and the dark sector all stand however the axis comes out. It remains the cheapest measurement that could light up a uniquely-ours signature, on archival plus near-future survey data (Euclid, Rubin/LSST). Reproducible as `figures/scripts/ch08_shared_axis.py`.

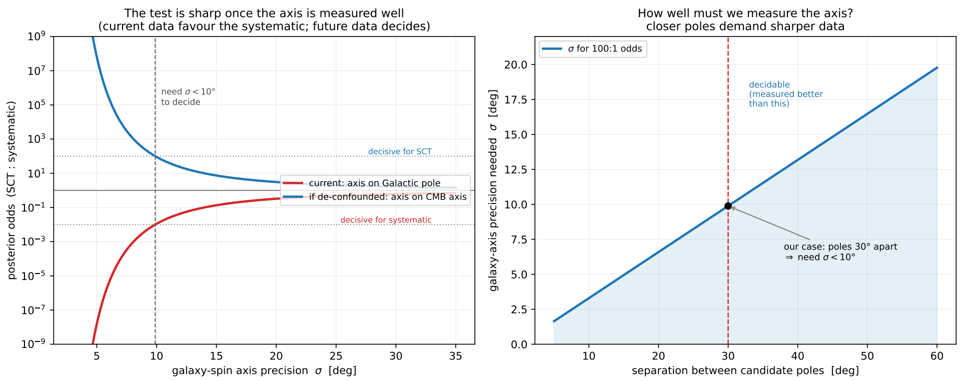

Figure 6.2: The shared-axis test as a precision question. *Left:* posterior odds (CMB axis : Galactic pole) versus the precision σ of the galaxy-spin axis; today's geometry (axis on the Galactic pole) favours the systematic, a de-confounded axis on the CMB axis would favour the framework, and either way $\sigma \lesssim 10^\circ$ is needed. *Right:* the candidate poles are $\sim 30^\circ$ apart, setting the $\sim 10^\circ$ precision target. Reproducible as `figures/scripts/ch08_shared_axis.py`.

Where we sit, and that it is typical. The same population statistics that make low spin typical let us locate our own footprint (Figure 6.3). Across three independent axes—generation depth (geometric, ∼90% of viable observers are first-generation children for branching $\epsilon \sim 0.1$), chirality (50/50 matter/antimatter by CPT, the halves observationally identical), and accretion family (clean and photon-dominated versus baryon-rich and degenerate)—our derived tags, a first-generation matter child of clean family at roughly a Hubble mass, land in the *dominant* cell, holding about 40% of all viable observers. That is the quantitative Copernican check: we are where most observers are, not a rare outlier. (The same accounting predicts *no* observers in the parentless first universes—dark-matter-free, they build almost no structure—nor in the baryon-rich degenerate tail; observers inhabit a narrow clean-family stripe, exactly our location. Reproducible as `ch08_census.py` and `ch08_universe_feel.py`.) This is a strong consistency test, though not by itself a discriminator against other cosmologies.

6.2 Dark energy that evolves

If dark energy is the parent's ongoing accretion (Chapter 5), it is a *rate*, generically not constant—so it should drift from a pure cosmological constant. DESI's baryon-acoustic-oscillation results favour exactly that: $w_0 w_a$ models with $w_0 > -1$ and $w_a < 0$ over strict Λ at the 2–3σ level [6], i.e. dark energy a little stronger than Λ today and weaker in the recent past. A parent-injection model—dark energy as the time-derivative of an accretion history that rose and then began to saturate—sits squarely in that (w_0, w_a) wedge, with a derived phantom crossing rather than a tuned sign (Figure 6.4). This is a statistical tension, not a confirmation: years 2–5 of DESI will sharpen it, and

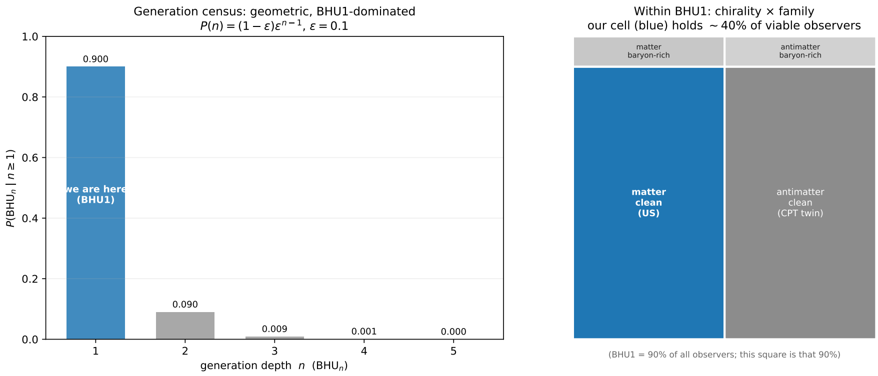

Figure 6.3: The supraverse census of viable observers. *Left:* the generation tower is geometric and first-generation-dominated; we are in the $n = 1$ bar. *Right:* within that generation, our cell (matter, clean family) holds ~40% of viable observers, with an identical antimatter *CPT* twin. Our footprint is the typical, dominant cell. Reproducible as `figures/scripts/ch08_census.py`.

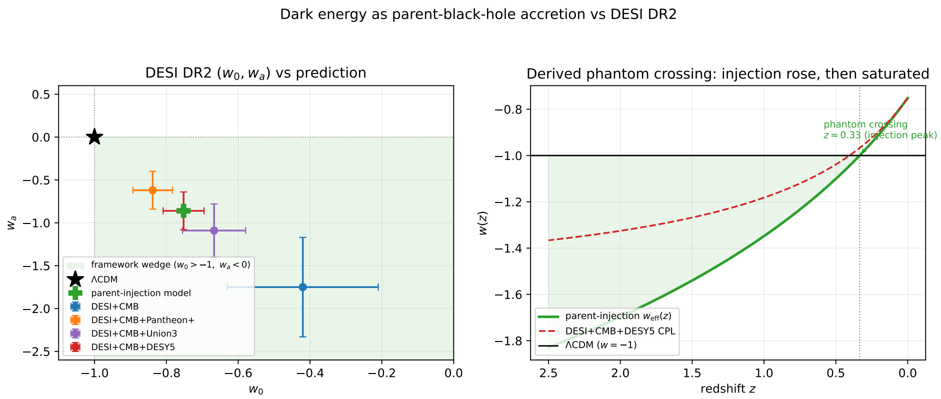

Figure 6.4: Dark energy as parent accretion versus DESI. *Left:* every DESI (w_0, w_a) combination sits in the framework's predicted wedge ($w_0 > -1$, $w_a < 0$), where the parent-injection model also lands. *Right:* the derived $w(z)$—a phantom crossing produced by accretion that rose then saturated, not a free sign choice. Reproducible as `figures/scripts/ch06_desi.py`.

if evolving dark energy is established at $> 5\sigma$, strict Λ is in trouble and accretion-driven dark energy of just this shape becomes the natural reading.

6.3 The Hubble tension

The $\sim 5\sigma$ gap between the early-universe Hubble value (CMB sound horizon, $H_0 \approx 67.4$) and the local distance ladder ($H_0 \approx 73.0$) costs the framework nothing to address, because it is natively inhomogeneous: voids and walls, with the dark sector a parent's accretion and the large-scale structure a void-dominated foam. It therefore inherits Wiltshire's *timescape* resolution [15, 16] with no added parameter. General relativity makes a void clock run faster

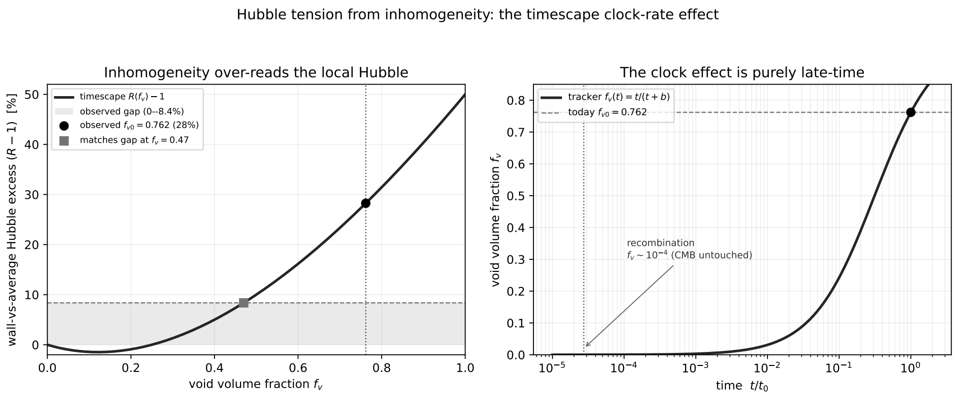

Figure 6.5: The Hubble tension from inhomogeneity. *Left:* the dressed/bare excess $R(f_v) - 1$—the factor by which a wall observer over-reads the volume-average Hubble—crosses the observed 8.4% gap (shaded) at $f_v \simeq 0.47$ and reaches 28% at the observed $f_{v0} \simeq 0.76$. *Right:* the tracker $f_v(t)$ is $\sim 10^{-4}$ at recombination (the CMB is untouched), rising to 0.76 today—the effect is purely late-time. Reproducible as `figures/scripts/ch08_hubble_timescape.py`.

than a wall clock, and we are wall observers, so the Hubble rate we infer—the *dressed* value—exceeds the volume-average *bare* value once voids fill the volume, by Wiltshire's closed-form factor $R(f_v) = (4f_v^2 + f_v + 4)/[2(2 + f_v)]$, with $R(0) = 1$. The tracker void fraction is $\sim 10^{-4}$ at recombination—so the early-universe value is *untouched*—and grows to the observed $f_{v0} \simeq 0.76$ today, where $R \simeq 1.28$: a 28% wall-versus-average excess, against an observed gap of only $\tau \simeq 8.4\%$ (Figure 6.5). The effect has the right sign, more than enough magnitude (the gap is matched already at $f_v \simeq 0.47$), and switches on only late—exactly the early-versus-late shape of the tension, controlled by the *measured* void fraction. This is a sign-and-magnitude result, not a global fit (the full timescape fit returns a single dressed $H_0 \simeq 61.7$); what it shows, parameter-free, is that inhomogeneity at the observed void fraction is amply capable of the tension. An eternal-Λ model has no such handle. Reproducible as `sims/src/cartasis_sims/hubble_timescape.py`.

6.4 Lumpy too early, and clustered too little

Two further tensions point the same way. JWST finds unexpectedly massive galaxies at $z > 10$, straining the standard structure-formation timeline; in this framework the inherited large-scale lumpiness of the parent's meal (Chapter 5) and a different post-bounce expansion give structure a head start, so early massive galaxies are less of a surprise—the great ultra-large structures (the Giant Arc, the Big Ring) are the same story on the largest scales. Meanwhile the σ_8 tension—low-redshift clustering weaker than the CMB-extrapolated prediction—is natural if dark matter is *geometric* (the parent's projected influence) rather than a clustering particle, since its effective late-time clustering need not match cold dark matter's. Both are live research questions, not yet quantitative predictions: each needs structure-formation modelling in the bounce framework (Chapter 8). They are flagged here as pointing in the framework's direction, not as won.

6.5 The matter excess, pinned twice

The most secure number in this chapter is the one the framework treats as *inherited*, not made (Chapter 4): the baryon-to-photon ratio $\eta \simeq 6 \times 10^{-10}$. It is worth stressing that the new reading of the CMB (the parent's Hawking glow, Chapter 5) does not put η in doubt, because η is anchored two independent ways that agree to about 1%: the CMB acoustic-peak ratio fixes the baryon density and hence $\eta_{10} \simeq 6.1$, and Big-Bang nucleosynthesis fixes η *separately* from the primordial deuterium abundance [5], also at $\eta_{10} \simeq 6.1$ (Figure 6.6, left). The Hawking-glow identification concerns only the *origin* of the thermal plasma at the bounce surface; it leaves recombination and nucleosynthesis untouched, so both legs stand and η is among the most secure numbers in cosmology.

Spin is local; chirality is inherited. Interpreting the axis test correctly requires keeping two properties apart (Figure 6.6, right). The *chirality* (the sign of η, matter versus antimatter) is inherited—set once at the founding seed and passed down a chirally pure lineage (Chapter 4). The *spin* (a universe's Kerr angular momentum and rotation axis) is *local*—it is the spin of our own progenitor black hole, formed astrophysically inside the parent and re-drawn every generation, not the distant founder's. So our observable preferred axis is our progenitor hole's, a single direction the CMB anomaly and the galaxy-spin handedness should share (§6.1)—while a quiet axis tells us only that our progenitor was low-spin, leaving our generation depth untouched. Reproducible as `figures/scripts/ch08_spin_eta.py`.

None of these is a confirmation on its own, and several are merely suggestive tensions awaiting better data or a calculation the framework still owes. But they are not nothing: an evolving dark energy of the predicted shape, a Hubble tension resolved without a new parameter, structure too lumpy for a smooth start, and a doubly-anchored matter excess are each what this cosmology expects and ΛCDM must accommodate. The single cleanest way to turn suggestion into evidence is the shared axis. The full tally—what is for, what is against, and what is still open—is drawn up in the scoreboard of Chapter 10.

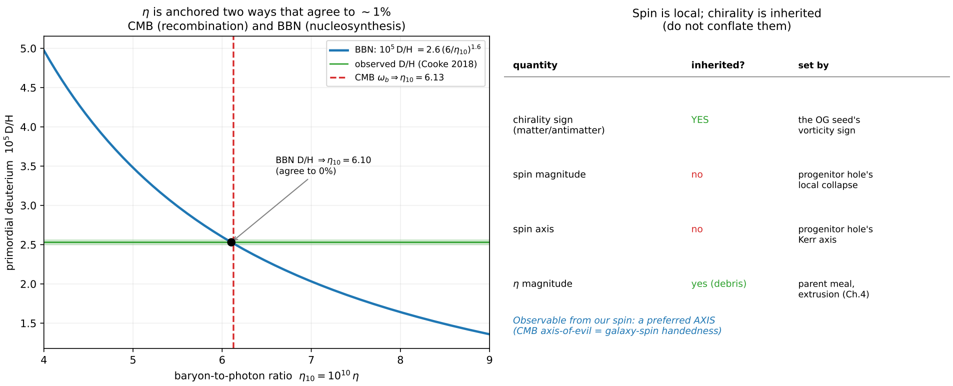

Figure 6.6: *Left:* η is anchored two independent ways that agree to $\sim$1%—the CMB-fixed baryon density and the BBN deuterium abundance—so the matter excess is secure despite the parent-Hawking reading of the CMB, which leaves nucleosynthesis untouched. *Right:* spin versus chirality— the chirality *sign* is inherited (pure lineages), while spin *magnitude and axis* are local (the progenitor hole's), so the observable is a single preferred axis. Reproducible as `figures/scripts/ch08_spin_eta.py`.

CHAPTER 7

What the Next Decade Will Decide

[Scaffold — content migrates here in Stage 4.]

CHAPTER 8

The Simulation Program

[Scaffold — content migrates here in Stage 4.]

CHAPTER 9

Seeing the Foam

[Scaffold — content migrates here in Stage 4.]

CHAPTER 10

The Scoreboard

[Scaffold — content migrates here in Stage 4.]

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